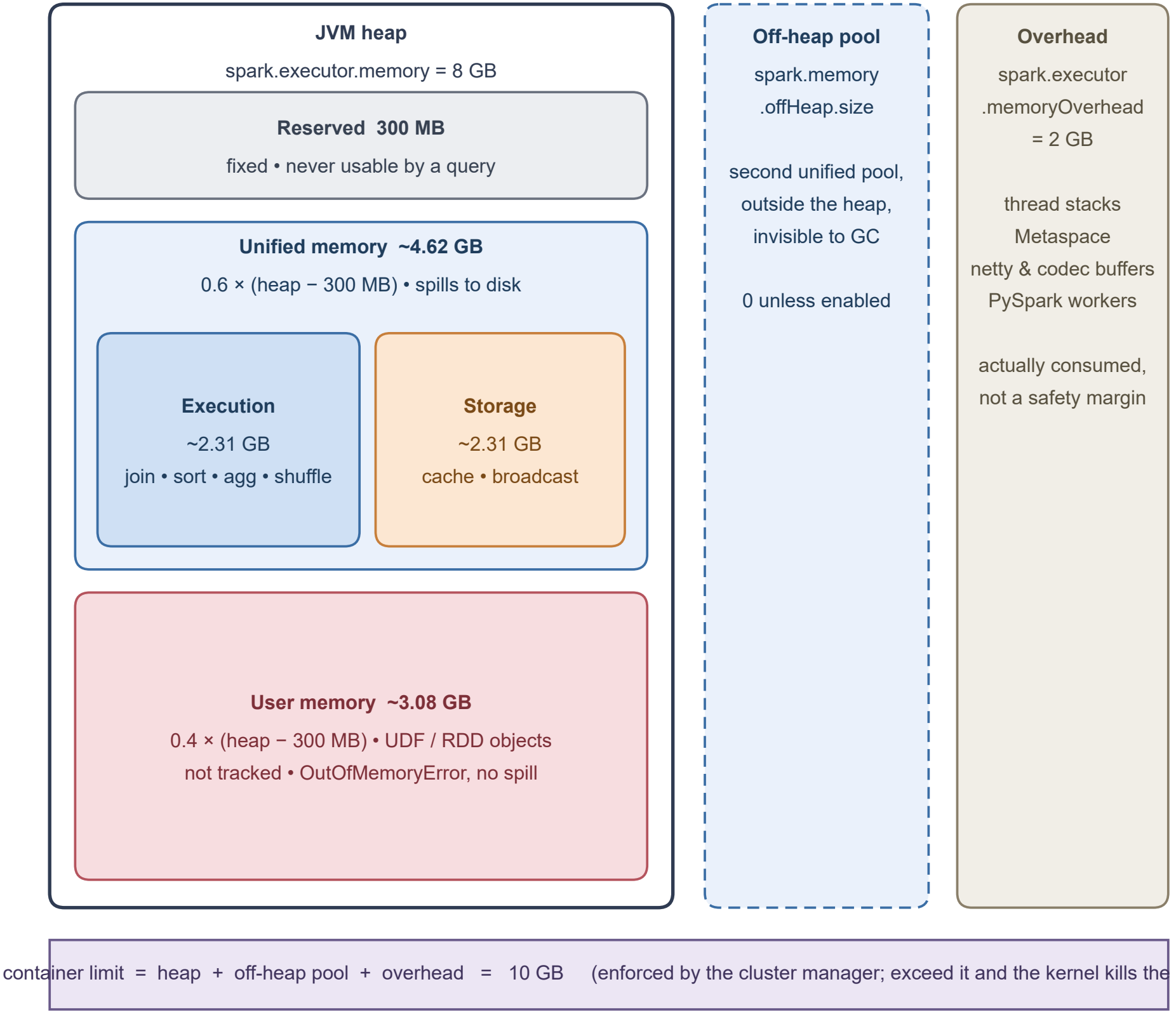

JVM heap
spark.executor.memory = 8 GB
Reserved 300 MB
fixed • never usable by a query
Unified memory ~4.62 GB
0.6 × (heap − 300 MB) • spills to disk
Execution
~2.31 GB
join • sort • agg • shuffle
Storage
~2.31 GB
cache • broadcast
User memory ~3.08 GB
0.4 × (heap − 300 MB) • UDF / RDD objects
not tracked • OutOfMemoryError, no spill
Off-heap pool
spark.memory
.offHeap.size
second unified pool,
outside the heap,
invisible to GC
0 unless enabled
Overhead
spark.executor
.memoryOverhead
= 2 GB
thread stacks
Metaspace
netty & codec buffers
PySpark workers
actually consumed,
not a safety margin
container limit = heap + off-heap pool + overhead = 10 GB (enforced by the cluster manager; exceed it and the kernel kills the pod)